

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
data = {
    "Student_ID": [1,2,3,4,5,6,7,8,9,10],
    "Age": [20, 21, 22, 23, np.nan, 21, 22, 100, 23, 21], # 100 = 100.0
    "Salary": [25000, 30000, 28000, 32000, 29000, 31000, 27000, 1000000, 23000, 30000],
    "City": ["Delhi", "Mumbai", "Delhi", "Chennai", "Mumbai", "Delhi", "Chennai", "Delhi", "Mumbai", "Mumbai"],
    "Gender": ["Male", "Female", "Male", "Female", "Male", "Male", "Female", "Male", "Female", "Male"]
}

df = pd.DataFrame(data)
df
```

	Student_ID	Age	Salary	City	Gender	
0	1	20.0	25000.0	Delhi	Male	
1	2	21.0	30000.0	Mumbai	Female	
2	3	22.0	28000.0	Delhi	Male	
3	4	23.0	32000.0	Chennai	Female	
4	5	NaN	29000.0	Mumbai	Male	
5	6	21.0	31000.0	Delhi	Male	
6	7	22.0	27000.0	None	Female	
7	8	100.0	1000000.0	Chennai	Male	
8	9	23.0	NaN	Delhi	Female	
9	10	21.0	30000.0	Mumbai	Male	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Shape & Structure of Dataset

```
df.shape
```

```
(10, 5)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
```

```
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Student_ID   10 non-null      int64
1   Age           9 non-null       float64
2   Salary        9 non-null       float64
3   City          9 non-null       object
4   Gender        10 non-null      object
dtypes: float64(2), int64(1), object(2)
memory usage: 532.0+ bytes
```

```
df.nunique()
```

```

      0
Student_ID  10
      Age     5
      Salary  8
      City    3
      Gender  2
```

```
dtype: int64
```

Basic Summary Statistics (Numerical Columns)

```
df.describe()
```

	Student_ID	Age	Salary
count	10.00000	9.000000	9.000000
mean	5.50000	30.333333	136888.888889
std	3.02765	26.143833	323673.618188
min	1.00000	20.000000	25000.000000
25%	3.25000	21.000000	28000.000000
50%	5.50000	22.000000	30000.000000
75%	7.75000	23.000000	31000.000000
max	10.00000	100.000000	1000000.000000

Summary Statistics for Categorical Columns

```
df.describe(include='object')
```

	City	Gender	
count	9	10	
unique	3	2	
top	Delhi	Male	
freq	4	6	

Individual Summary Statistics (Column-wise)

```
df['Salary'].min()
df['Salary'].max()
```

1000000.0

Mean (Average)

```
df['Salary'].mean()
```

np.float64(136888.88888888888)

```
df['Salary'].median()
```

30000.0

```
df['City'].mode()
```

City

0 Delhi

dtype: object

Exhaustive Statistics for Each Numerical Feature

```
numerical_df = df.select_dtypes(include=[np.number])

for column in numerical_df.columns:
    print(f"\nFeature: {column}")
    print("Minimum Value      :", numerical_df[column].min())
    print("Maximum Value       :", numerical_df[column].max())
    print("Mean                  :", numerical_df[column].mean())
    print("Median                 :", numerical_df[column].median())
```

```

print("Range      :", numerical_df[column].max() - numerical_df[column].min())
print("Standard Deviation:", numerical_df[column].std())
print("Variance      :", numerical_df[column].var())
print("25th Percentile  :", numerical_df[column].quantile(0.25))
print("50th Percentile  :", numerical_df[column].quantile(0.50))
print("75th Percentile  :", numerical_df[column].quantile(0.75))

```

```

Feature: Student_ID
Minimum Value      : 1
Maximum Value      : 10
Mean               : 5.5
Median             : 5.5
Range              : 9
Standard Deviation: 3.0276503540974917
Variance           : 9.166666666666666
25th Percentile    : 3.25
50th Percentile    : 5.5
75th Percentile    : 7.75

```

```

Feature: Age
Minimum Value      : 20.0
Maximum Value      : 100.0
Mean               : 30.333333333333332
Median             : 22.0
Range              : 80.0
Standard Deviation: 26.143832924802744
Variance           : 683.5
25th Percentile    : 21.0
50th Percentile    : 22.0
75th Percentile    : 23.0

```

```

Feature: Salary
Minimum Value      : 25000.0
Maximum Value      : 1000000.0
Mean               : 136888.88888888888
Median             : 30000.0
Range              : 975000.0
Standard Deviation: 323673.61818830884
Variance           : 104764611111.11111
25th Percentile    : 28000.0
50th Percentile    : 30000.0
75th Percentile    : 31000.0

```

NumPy Summary Statistics

```

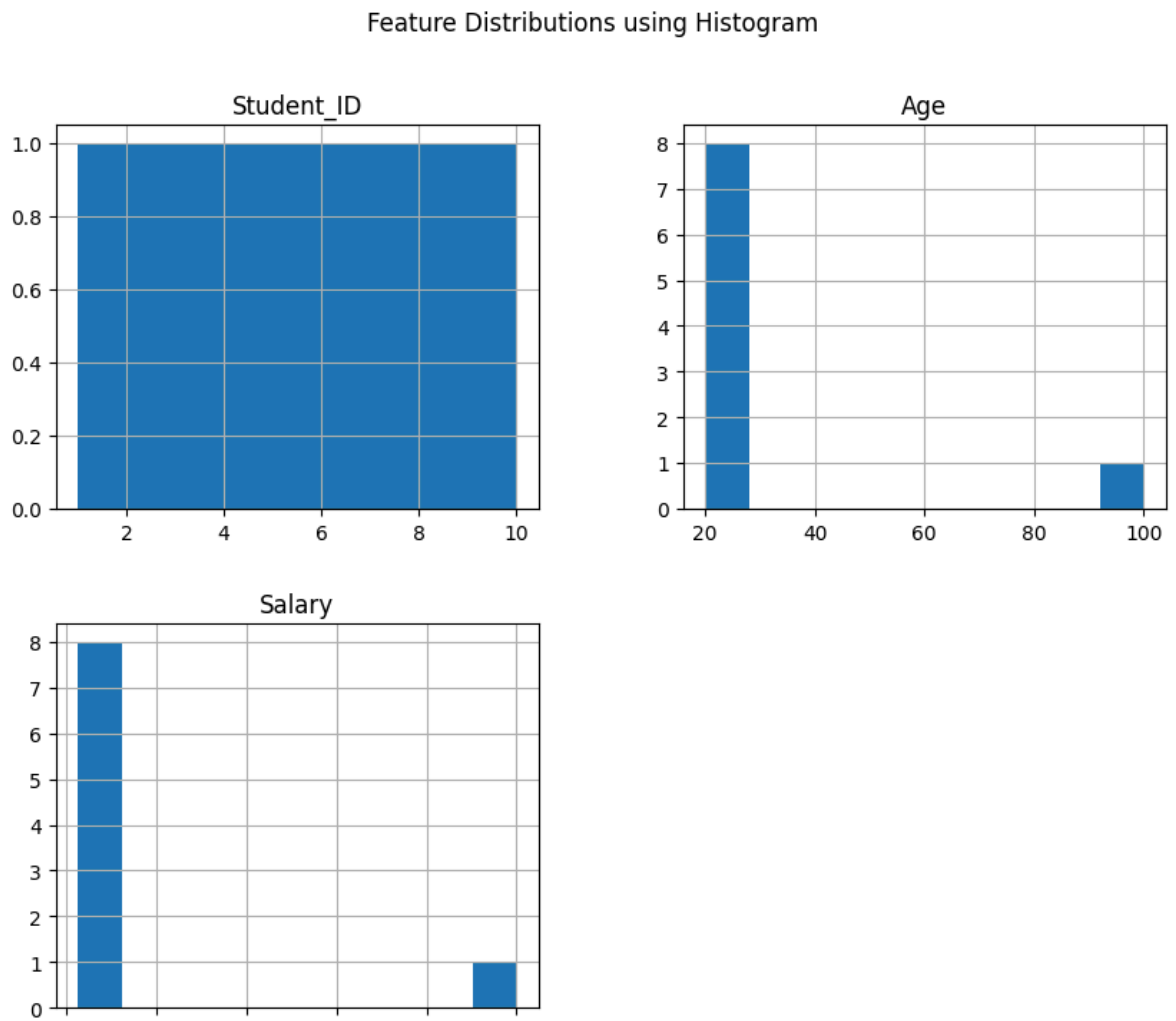
np.mean(df['Salary'])
np.median(df['Salary'])
np.std(df['Salary'])
np.var(df['Salary'])

```

93124098765.4321

Histogram

```
numerical_df.hist(bins=10, figsize=(10,8))  
plt.suptitle("Feature Distributions using Histogram")  
plt.show()
```



Remove Duplicate Records

```
df.drop_duplicates(inplace=True)
```

Handle Outliers

```
Q1 = df['Salary'].quantile(0.25)  
Q3 = df['Salary'].quantile(0.75)  
IQR = Q3 - Q1  
  
df = df[(df['Salary'] >= Q1 - 1.5 * IQR) &  
        (df['Salary'] <= Q3 + 1.5 * IQR)]
```

Create Second Dataset

```
marks = pd.DataFrame({
    "Student_ID": [1,2,3,4,5,6,7,8],
    "Marks": [78, 85, 69, 90, 88, 76, 95, 60]
})
```

marks

	Student_ID	Marks	
0	1	78	
1	2	85	
2	3	69	
3	4	90	
4	5	88	
5	6	76	
6	7	95	
7	8	60	

Next steps:

[Generate code with marks](#)
[New interactive sheet](#)

Inner Join

```
merged_inner = pd.merge(df, marks, on="Student_ID", how="inner")
merged_inner
```

	Student_ID	Age	Salary	City	Gender	Marks	
0	1	0.000000	-1.763834	Delhi	Male	78	
1	2	0.333333	0.440959	Mumbai	Female	85	
2	3	0.666667	-0.440959	Delhi	Male	69	
3	4	1.000000	1.322876	Chennai	Female	90	
4	5	NaN	0.000000	Mumbai	Male	88	
5	6	0.333333	0.881917	Delhi	Male	76	
6	7	0.666667	-0.881917	None	Female	95	

Next steps:

[Generate code with merged_inner](#)
[New interactive sheet](#)

Left Join

```
merged_left = pd.merge(df, marks, on="Student_ID", how="left")
```

Save Final Cleaned & Integrated Dataset

```
merged_inner.to_csv("final_dataset.csv", index=False)
```